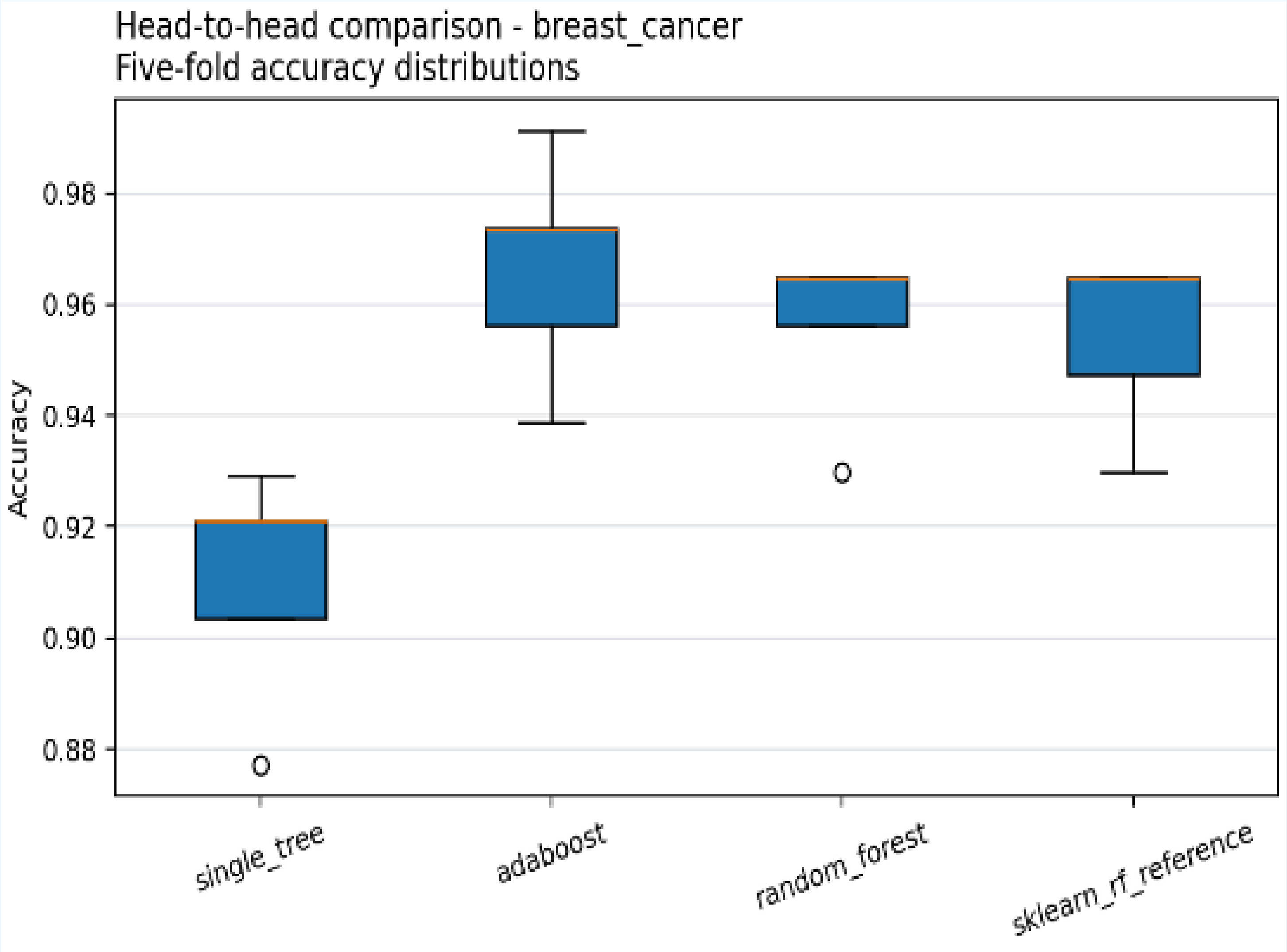
# When does boosting beat bagging?

A from-scratch study of bias, variance, label noise, imbalance, and data structure




Member A | Member B | Member C | Member D

AI Academy | Spring 2026

# The methods attack different errors

BOOSTING

## Correct

**Fit a stump**

**Raise hard-example weight**

Sequential attention can lower residual boundary bias  and can chase corrupted labels.

BAGGING

## Average

**Bootstrap**

**Decorrelate trees**

Independent variation cancels; shared correlation remains. OOB rows provide an internal check.

# Inspectable code - not a library wrapper

## 7

**from-scratch algorithms**

Tree | SAMME | Forest
PCA | K-Means | DBSCAN
Gradient Boosting

## 51

**passing tests; 88.71% coverage**

Weighted splits | OOB complements
Empty clusters | noise-to-border
Sequential/parallel identity

## 1

**binding reproduction command**

python src/experiments/run_all.py
Seed 42 | manifests | raw fold/stage/replicate evidence

# Three datasets isolate the conditions that should change the winner

DATA

| | |
|---|---|
| **Breast Cancer** | 569 rows \| 30 features \| balanced binary baseline and Brier study |
| **Adult Income** | 48,842 rows \| 14 raw features \| mixed types, missingness, moderate imbalance |
| **Covertype** | 10,000-row subset \| 54 features \| 7 classes \| minority 0.47% |

SEVEN REQUIRED EXPERIMENTS

1 Baseline and sklearn tolerance | 2 AdaBoost stages: 1-200 | 3 Forest size and depth | 4 Identical five-fold head-to-head

5 Nested label-noise replicates | 6 100 shared bootstrap fits | 7 PCA, K-Means, DBSCAN

**Split first. Fit preprocessing and oversampling inside training only.**

# Custom trees stay within reference tolerance on every dataset

**0.912**

**custom tree accuracy**

**0.912**

**sklearn tree accuracy**

**0.8 pp**

maximum gap across datasets | required <= 2.0 pp

### Why the match is credible

Sorted midpoint search | weighted impurity | deterministic ties | identical train/test split

### Why agreement is not identity

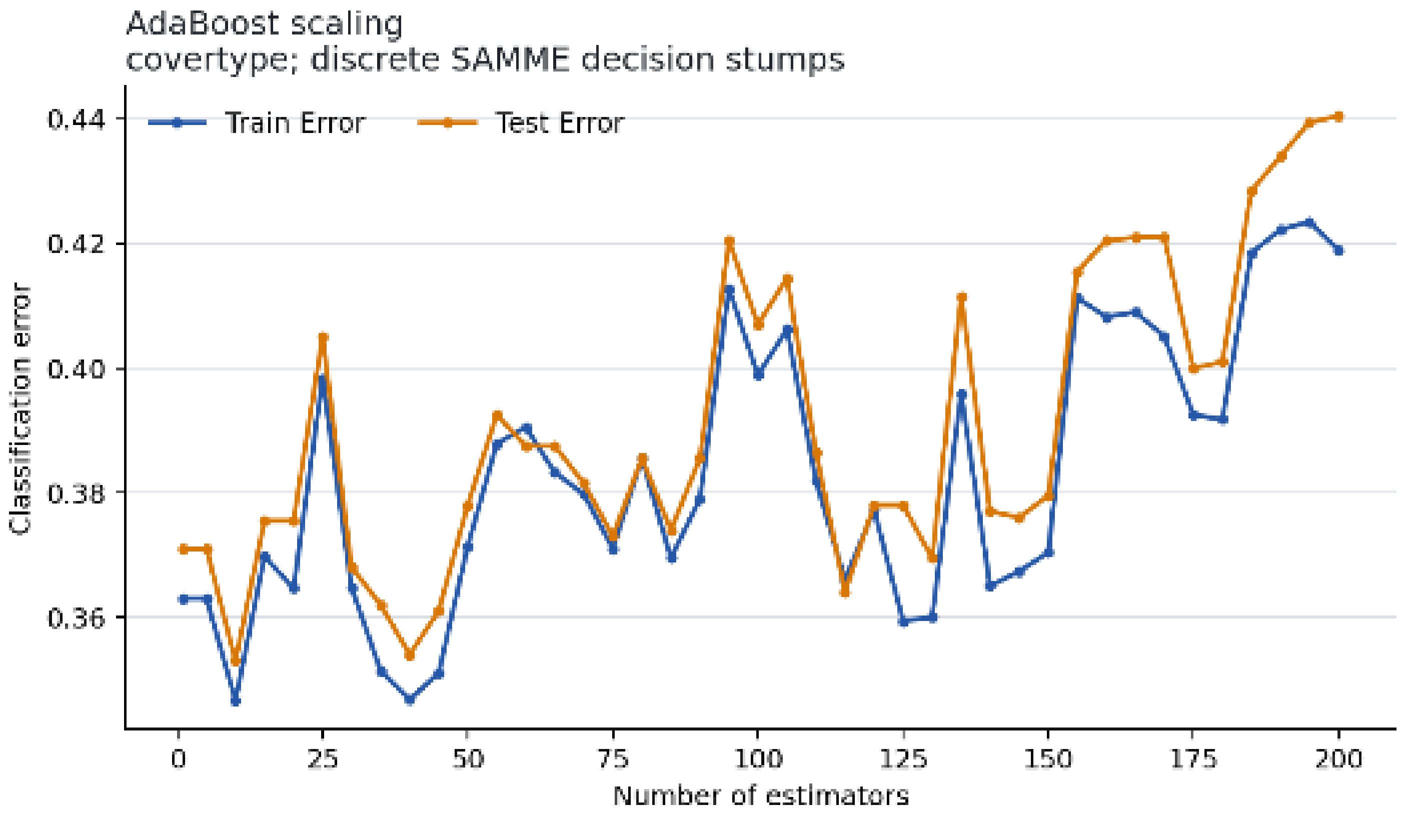
Different tie handling can change tree shape; tests and invariants remain necessary.

### Acceptance

**PASS**

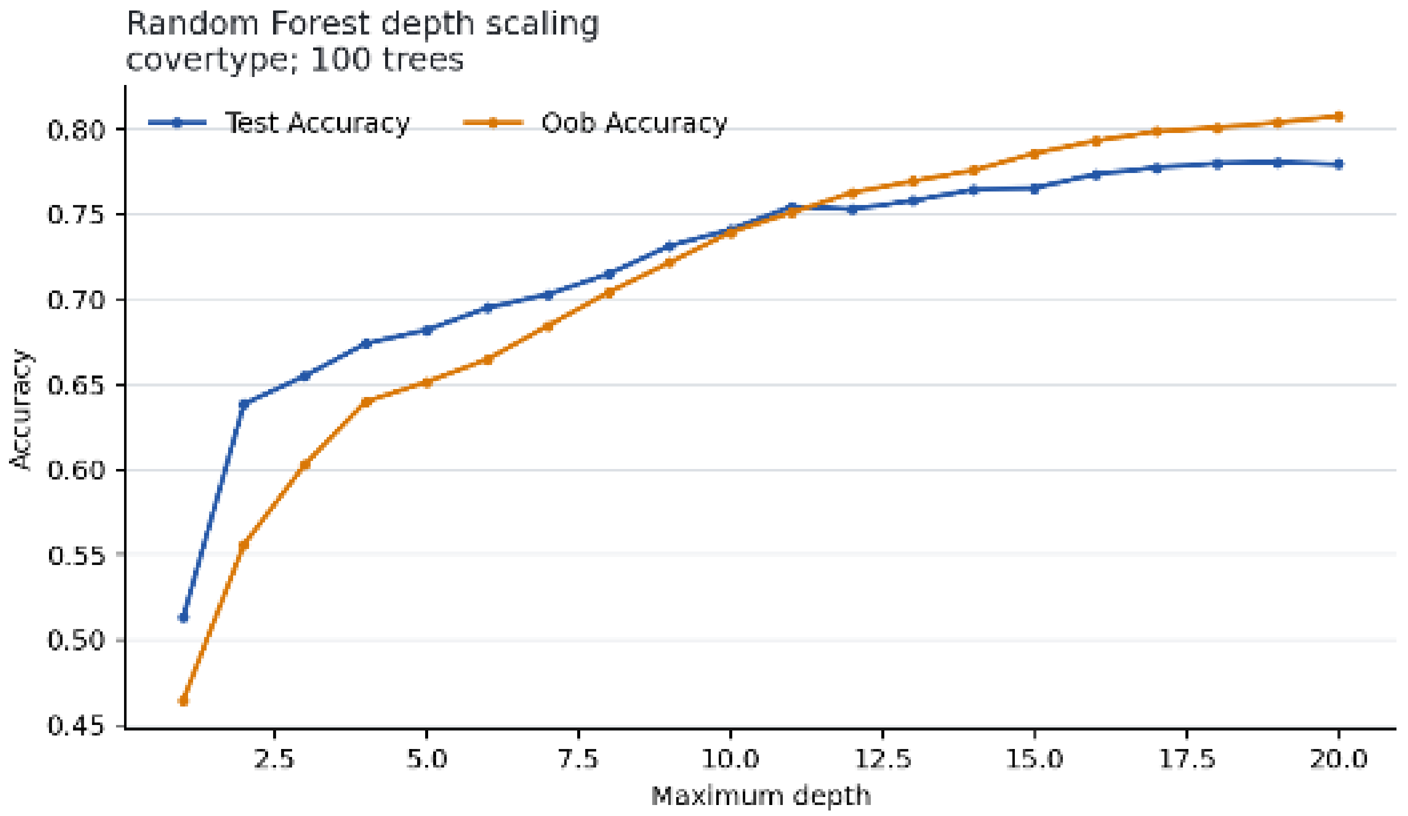
# Scaling exposes different complexity needs

### AdaBoost: 1-200



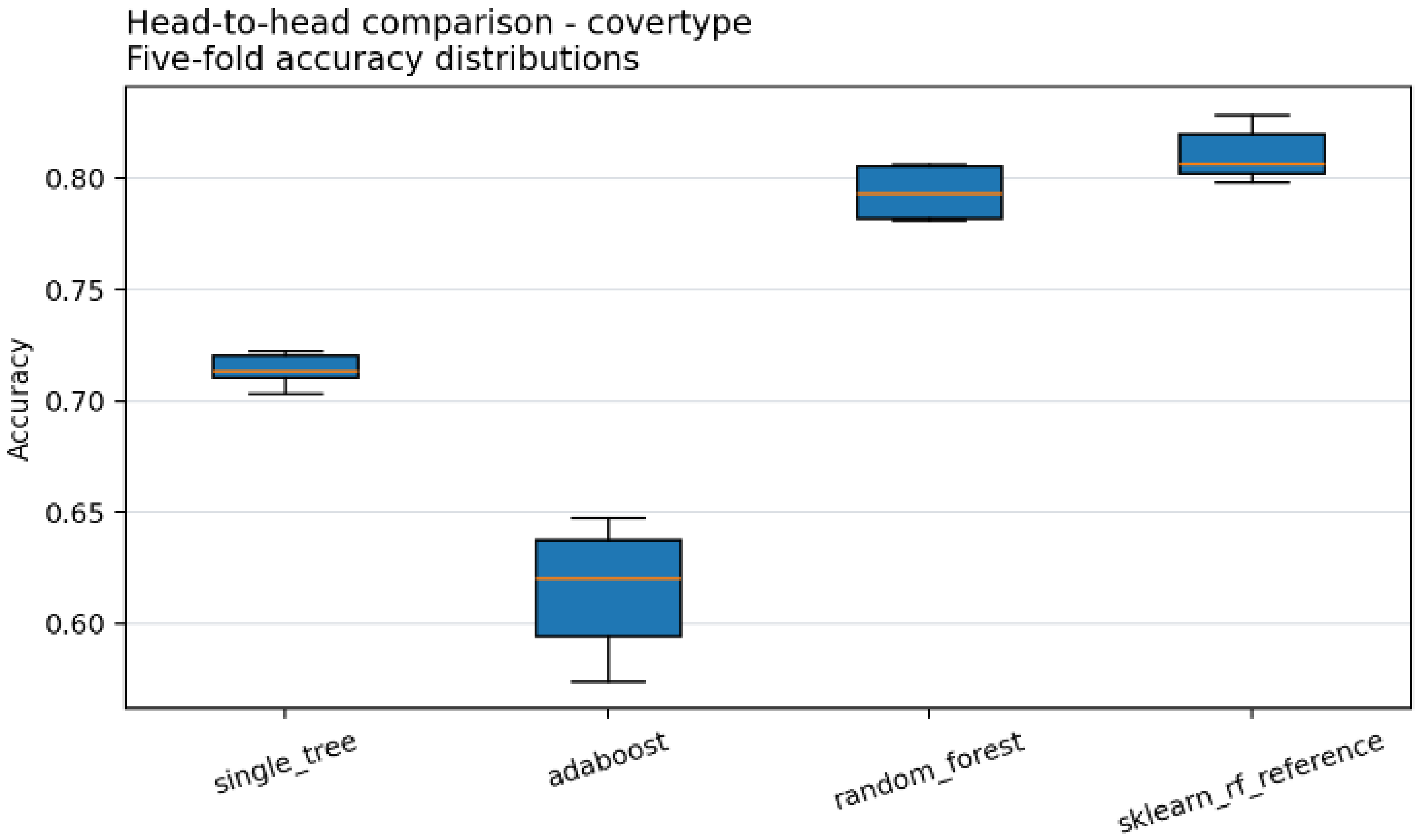

**Best error at 10; later rounds hurt**

### Forest: depth 1-20




**Best depth 19; flexible trees matter**

# The winner changes with dataset structure




**FIVE-FOLD MEAN ACCURACY**

## .9666

Breast Cancer Ada | RF .9561

## .8649

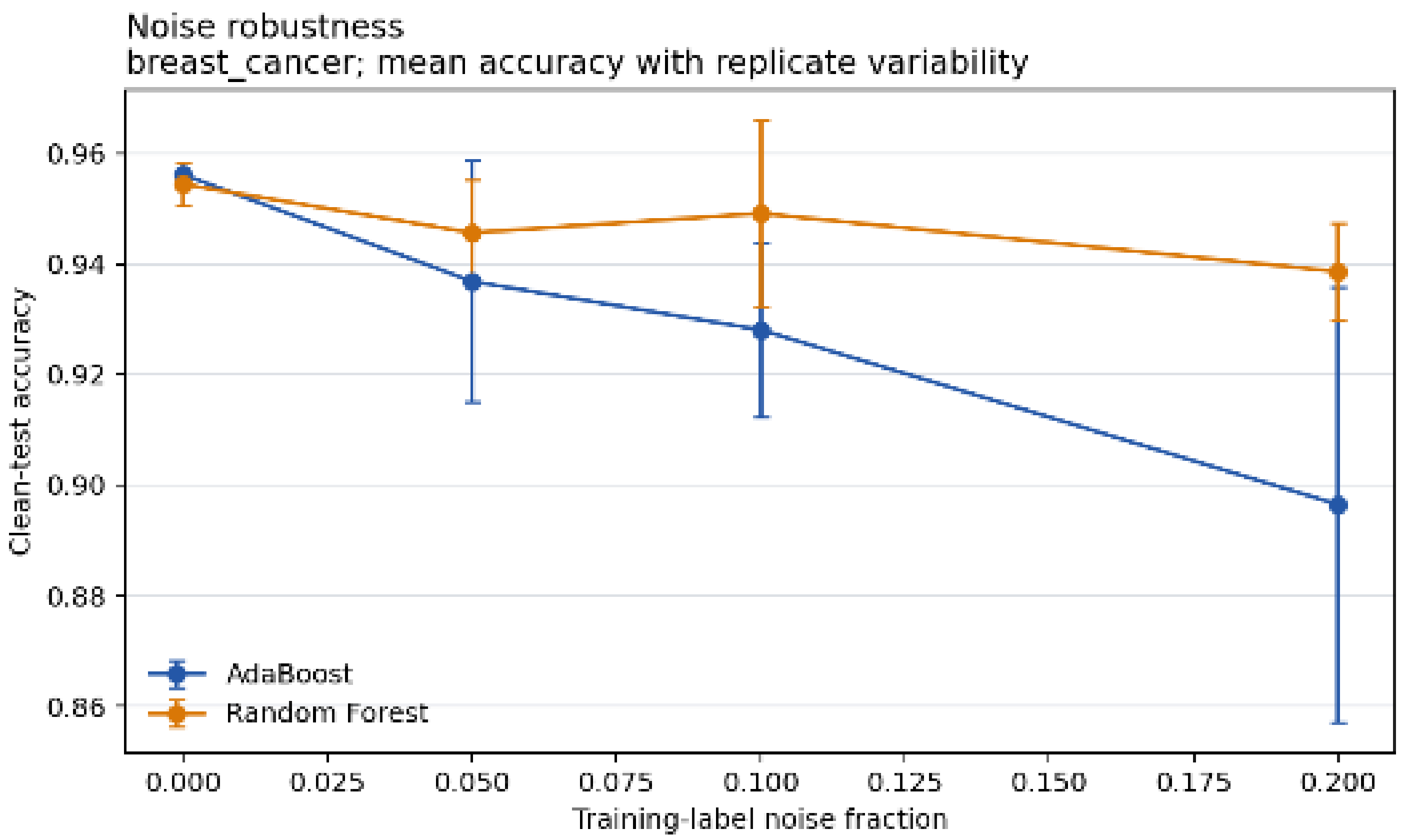
Adult RF | Ada .8556

## .7938

Covertype RF | Ada .6150

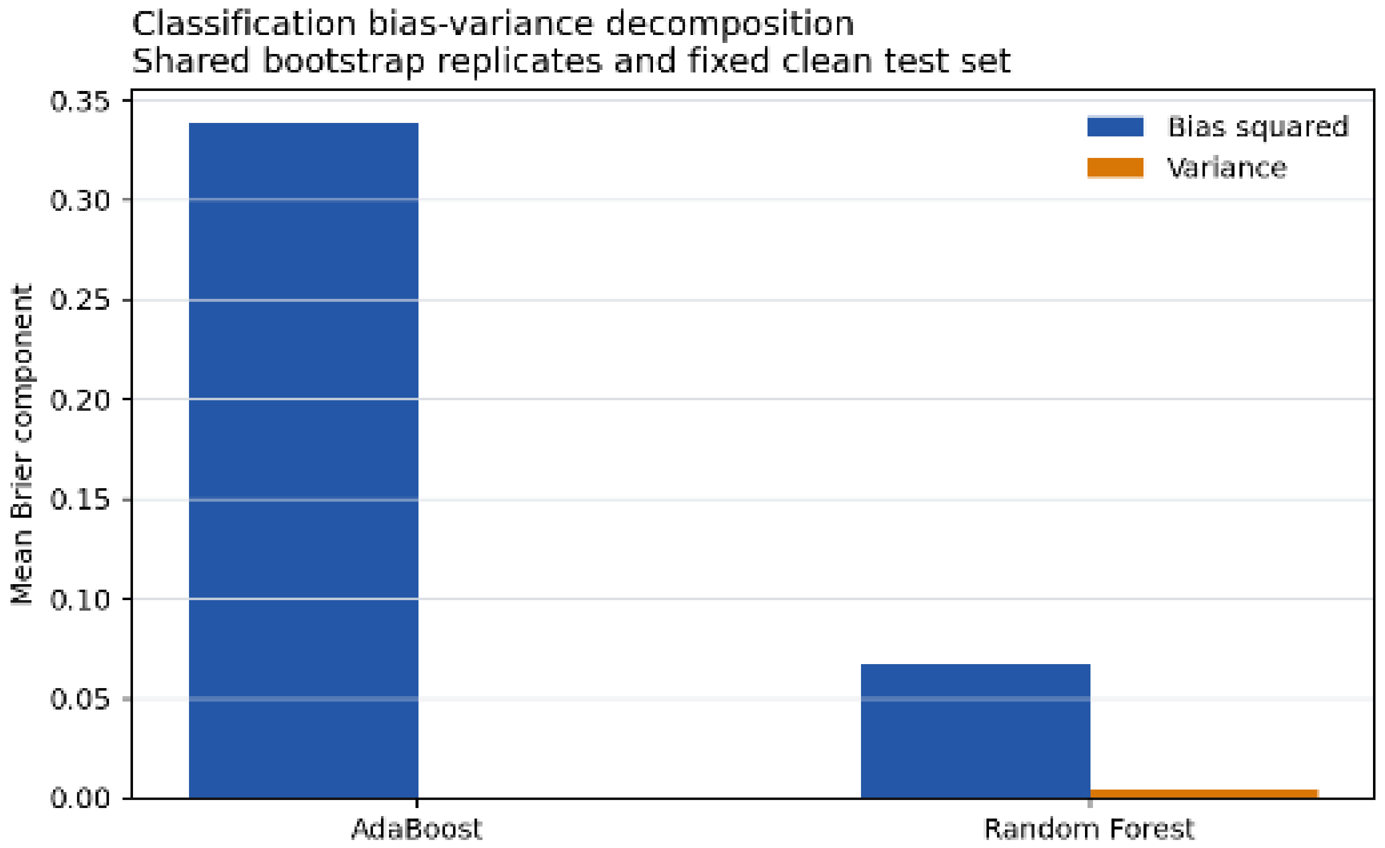
# Noise robustness and probability loss tell different stories

**20% LABEL NOISE | 5 REPLICATES**



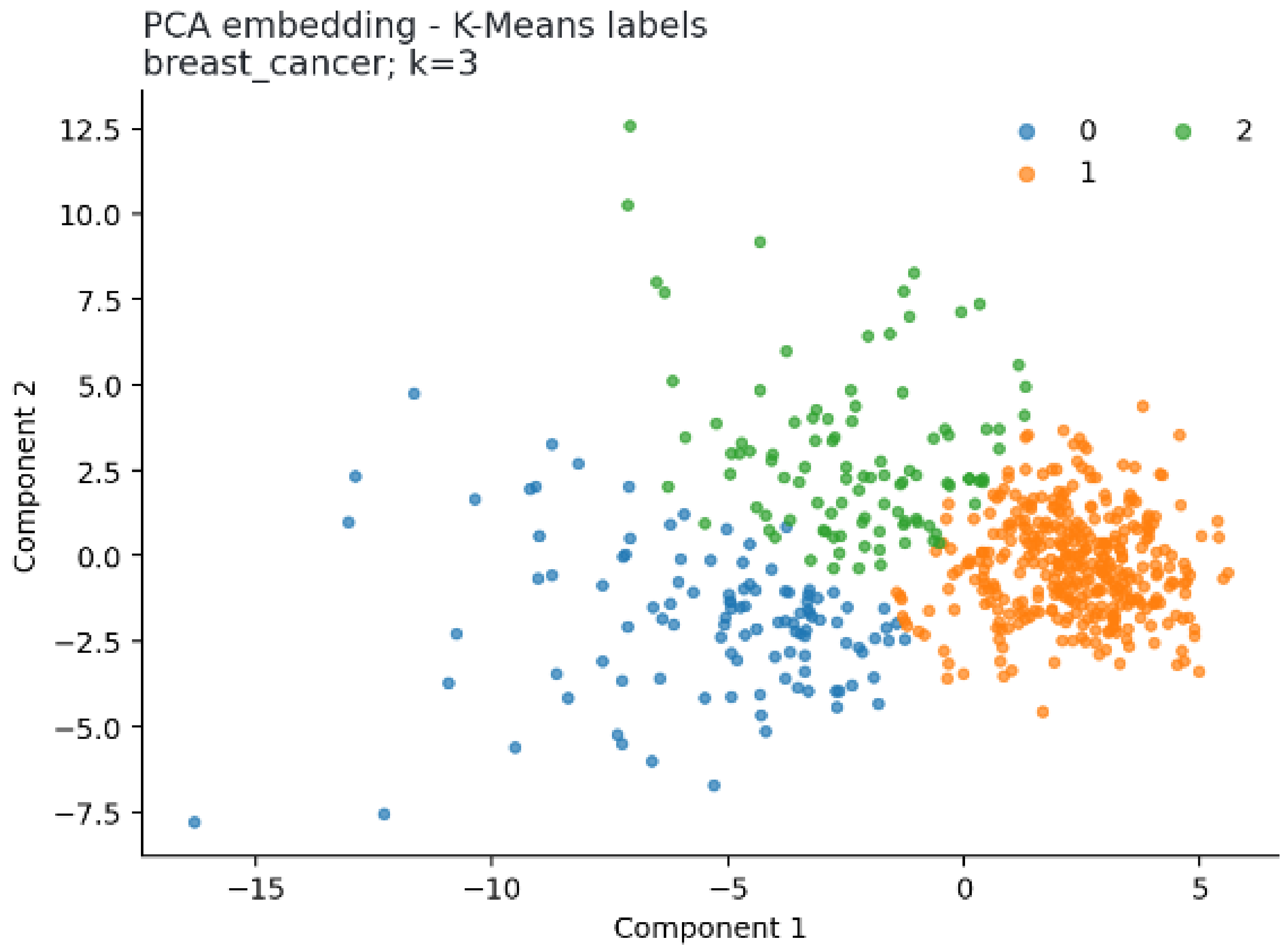

At 20%: AdaBoost -5.96 pp; forest -1.58 pp.

**BRIER | 100 BOOTSTRAPS**




Bias^2: AdaBoost .3382; forest .0675. Votes are not calibrated.

# Clusters reveal geometry, not guaranteed class boundaries




**7** PCs reach >= 90% variance

**.515** K-Means ARI | k = 3

**6.50%** DBSCAN noise | ARI = .030

Full dataset n = 569

# Bonuses extend the analysis without replacing it

### SAMME.R

BC .9561 to .9649
Adult .8573 to .8687

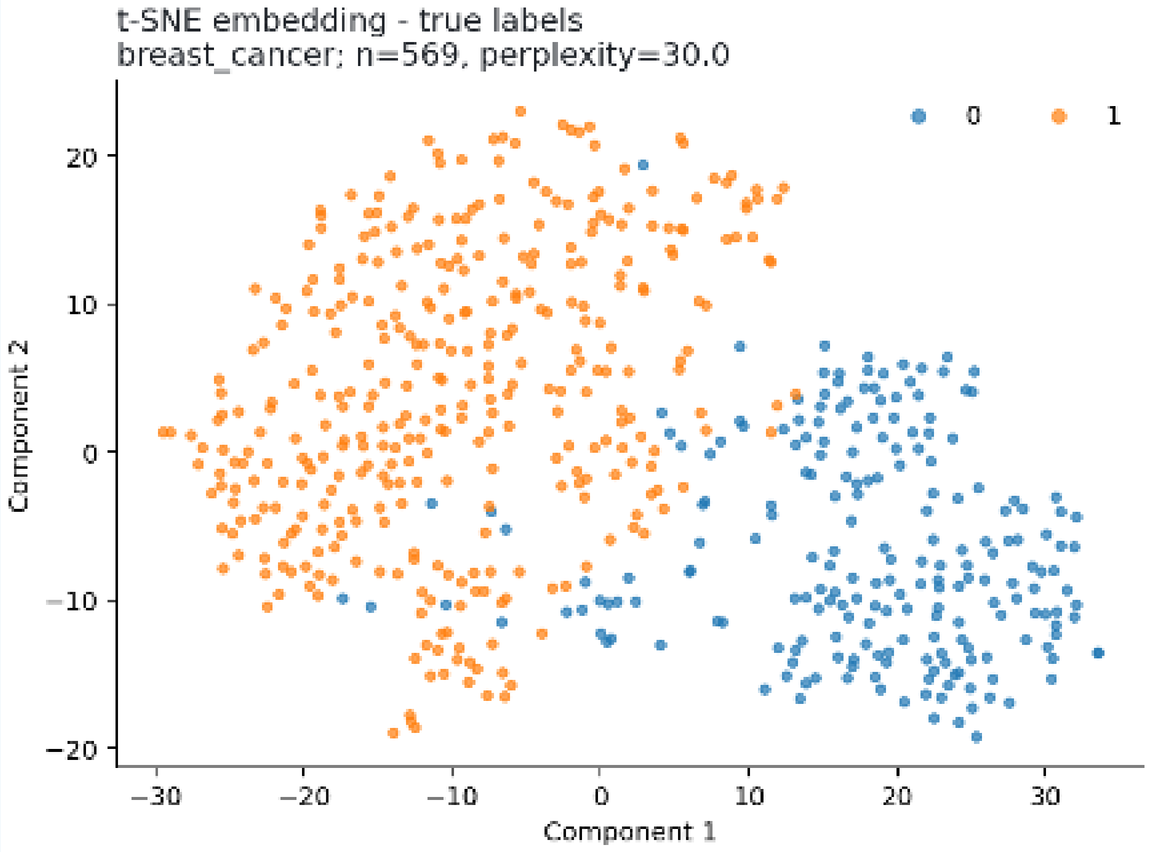
### Gradient Boosting

Binary log-loss; BC accuracy .9386, AUC .9899

### Interactive notebook

Executed controls for model and cluster complexity

### t-SNE




n=569, perplexity 30 | no global-distance claim

**88.71% coverage | Ruff clean | mypy clean | AI assistance disclosed**

THE ANSWER

# No universal winner: let the data decide

### BOOST

Breast Cancer: AdaBoost .9666. Sequential correction worked best on this structured binary task.

### BAG

Adult RF .8649; Covertype RF .7938. Diverse trees handled mixed and multiclass structure better.

**Release after truthful contribution approval, genuine Git history, final audit, Moodle submission, and v1.0-final.**